

LENDING

The complete loan lifecycle.

From onboarding and credit assessment through issuance, scheduling, recovery, and renewal — how FinFlo runs lending end-to-end, with bank-grade money handling at every step.

DOCUMENT

Loan Management

AUDIENCE

Operations · Lending teams

VERSION

1.0

PRECISION ENGINEERING FOR MODERN FINANCE

LIFECYCLE AT A GLANCE

| One continuous flow, from application to closure

Every loan in FinFlo travels the same disciplined path. Each stage hands a clean, auditable state to the next, so balances, repayment ledgers, and reports always reconcile — and the heavy money work is automated by scheduled jobs rather than left to manual reconciliation.



Onboarding & KYC

The customer is onboarded with CNIC identity capture and document KYC. Sensitive personal fields are encrypted at rest.



Credit assessment

A credit limit is resolved from the member's share balance and a credit-score band; an optional risk grade and score support the decision.



Issuance

On approval the loan is created with its principal, rate, duration, EMI, total amount, and start date, and the principal is disbursed.



Repayment schedule

Installments are tracked through the Repayment ledger; each payment splits into principal and interest portions.



Recovery

Daily jobs flag overdue installments, accrue late fees, and detect defaults — all in the Asia/Karachi timezone.



Closure or renewal

A fully repaid loan completes; an eligible customer can roll over, top up, or extend their borrowing through a renewal.

Why a single flow matters

Because onboarding through payout lives in one system, the loan's principal, accrued interest, late fees, and outstanding balance are never reconstructed from disconnected tools — every number traces back to a ledger row.

LOAN PRODUCTS & TERMS

| The fields that define a loan

A FinFlo loan is described by a small, explicit set of terms. **Simple interest** is the default interest type, with EMI-style and compound modes also supported. Money fields are rounded to whole rupees at rest so a fractional rupee can never be persisted.

FIELD	WHAT IT REPRESENTS
Principal	The amount lent — the capital the customer receives.
Rate	The interest / profit rate applied, as a percentage.
Duration	The loan term, expressed in months.
Interest type	<code>simple</code> (default), <code>emi</code> , or <code>compound</code> .
EMI	The monthly installment the customer pays.
Total amount	Total repayment due over the term — principal plus interest.
Start date	When the schedule begins; the maturity follows from the duration.
Outstanding principal	Principal still owed, tracked separately from the blended remaining balance so interest accrues on principal only — never on fees or already-capitalized interest.

Illustrative example — repayment schedule

A PKR 120,000 principal over 12 months at a simple rate produces a fixed monthly EMI; each payment is split into an interest portion and a principal portion, and the outstanding principal steps down month by month until the total amount is fully repaid. *Figures are an illustrative example, not an official product or rate.*

CREDIT LIMITS & ELIGIBILITY

| Capacity anchored to capital, modulated by behavior

A member's borrowing capacity is computed by the `creditLimitService`. Capacity stays anchored to the member's **share balance** (their business-share investment), and a **credit-score band** raises or lowers it — strong repayment history visibly increases the limit, poor history reduces it. A non-member customer has no share balance, so no automatic limit is enforced for them.

Share-anchored base

The base limit is derived from the member's share balance, then scaled by their score band (Excellent through Very Poor).

KYC completeness

Identity and document KYC must be in place before a member can borrow; sensitive fields are stored encrypted.

Per-tenant scoping

Every limit, member, and loan is scoped to its business owner, so one tenant's lending is fully isolated from another's.

Credit scoring must never break limit resolution — if a score cannot be computed, the service falls back to a neutral band rather than failing.

REPAYMENT & SCHEDULING

| Every payment is a ledger row

Repayments are recorded as **Repayment** documents linked to the loan, customer, and tenant. Each row carries the total amount paid, its split into `principalAmount` and `interestAmount`, the installment number, and a date. As payments land, the loan's paid amount rises, its remaining balance and outstanding principal fall, and the loan completes when the full total is repaid.

Installment schedule

Payments are tracked against numbered installments, so the system always knows which periods are settled and which are still due.

Principal / interest split

Each repayment separates principal from interest, keeping the outstanding-principal figure exact for accrual.

Partial & early payments

The ledger accepts partial and early payments — the balance moves by exactly what was paid, with no fractional rupee left behind.

On-time vs late

Whether a payment is on time or late is determined against the installment's due date, which drives downstream overdue and late-fee handling.

Repayment posting is a multi-document money operation, so it runs inside a database transaction and writes an auditable ledger entry — see "Money safety" below.

LATE FEES

| Automated accrual, never double-charged

Late fees are applied by the `lateFeeService`, driven by a **daily late-fee accrual job** (`runLateFeeAccrual`). A separate **overdue downgrade job** (`runOverdueDowngrade`) moves loans that have missed their due date into the overdue state each day. Both run on a fixed daily schedule in the **Asia/Karachi** timezone.

One fee per loan per period

A loan records which engine applied its late fee — the daily cron accrual or a manual flat fee. Whichever touches a loan first in a calendar month "owns" it for that month and the other engine skips it, so the two can never stack a fee on the same loan in the same period.

Late-fee accrual is governed by per-tenant settings (whether late fees are enabled, the fee type and rate, and a grace period), so each business applies its own recovery policy.

DEFAULT DETECTION

| Status transitions, decided by a daily job

A scheduled **default-detection job** (`runLoanDefaultDetection`) runs daily in the Asia/Karachi timezone and advances loans that have fallen far enough behind into the defaulted state, recording when and why. A loan's lifecycle is a small, explicit set of statuses:

STATUS	MEANING
pending	Awaiting approval — not yet disbursed.
active	Disbursed and being repaid on schedule.
overdue	A due installment was missed; flagged by the daily downgrade job.
defaulted	Sufficiently delinquent; set by the default-detection job with a reason and timestamp.
completed	Fully repaid — the loan is closed.
renewed	Continued through a renewal into a successor loan or extended in place.
rejected	Declined at approval, with a recorded reason.

RENEWALS

Continuing a customer's borrowing

When a customer is ready to keep borrowing, FinFlo supports three renewal modes, each with explicit linkage between the old and new records so history is never lost.



Rollover

The existing loan is replaced by a fresh successor loan; the old loan points forward and the new one points back to it.



Top-up

The customer's borrowing is increased into a new loan that continues from the prior one.



Extend

No new record is created — the same loan is mutated in place, and its renewal count and last-renewed date are bumped.

Renewal linkage (`renewedFrom` / `renewedTo`) keeps the full chain of a customer's borrowing traceable across cycles.

GROUP LENDING

Joint-liability groups, individual sub-loans

For the classic microfinance methodology, FinFlo models a standing **LoanGroup** — a joint-liability group of borrowers with a leader (the informational point of contact) and members, plus a guarantee policy. Lending happens through **GroupLoan** cycles: each cycle fans out to **one individual loan per member**, sized to that member's request (per-member allocations).

The group is the collateral

Under a `joint` guarantee policy, if any member's sub-loan goes overdue or defaults the whole group is flagged `at_risk` and new group lending is frozen until it recovers.

The engine is unchanged

Each sub-loan is an ordinary loan. The repayment service, late-fee and default jobs, and ledger all run on it as usual — the GroupLoan only ties them together and keeps denormalized roll-ups in sync.

Aligned schedules

Shared terms (rate, duration, interest type, start date) are applied to every sub-loan so installment dates line up for group-meeting collection.

Repeat cycles

A group that clears or rolls over one cycle can borrow again as the next cycle, with the same renewal linkage as individual loans.

Money safety — non-negotiable at every step

Every change to a balance, repayment, distribution, or transfer runs inside a MongoDB session/transaction with idempotent patterns (guarded `findOneAndUpdate` , `$inc`), and an auditable `FinancialTransaction` ledger row is written for the money movement. `abortTransaction()` is called on every early-return and error path, so a failed operation rolls back cleanly and never leaks an open transaction that would hold locks. All money logic lives in services, not controllers.

